Research Note 4

1 $\chi\bar{\chi} \to f\bar{f}$ check

$\chi\bar{\chi} \to f\bar{f}$ and $\chi\bar{\chi} \to \nu\bar{\nu}$, $|\mathcal{M}|^2$ `FeynCalc`, initial state average, final state sum

1.1 code for $|M|^2$

```
ClearAll["Global`*"];
Needs["FeynCalc`"];

denS[s_] := s - MA^2;
denT[t_] := t - MA^2;
denU[u_] := u - MA^2;

GetMandelstamRules[s_, t_, u_, m1_, m2_, m3_, m4_] :=
  FCI[{SP[p1, p1] -> m1^2, SP[p2, p2] -> m2^2, SP[p3, p3] -> m3^2,
    SP[p4, p4] -> m4^2, SP[p1, p2] -> (s - m1^2 - m2^2)/2,
    SP[p2, p1] -> (s - m1^2 - m2^2)/2,
    SP[p3, p4] -> (s - m3^2 - m4^2)/2,
    SP[p4, p3] -> (s - m3^2 - m4^2)/2,
    SP[p1, p3] -> (m1^2 + m3^2 - t)/2,
    SP[p3, p1] -> (m1^2 + m3^2 - t)/2,
    SP[p2, p4] -> (m2^2 + m4^2 - t)/2,
    SP[p4, p2] -> (m2^2 + m4^2 - t)/2,
    SP[p1, p4] -> (m1^2 + m4^2 - u)/2,
    SP[p4, p1] -> (m1^2 + m4^2 - u)/2,
    SP[p2, p3] -> (m2^2 + m3^2 - u)/2,
    SP[p3, p2] -> (m2^2 + m3^2 - u)/2}];

ProcessCleanUp[rawMatrix_, rule_, elimRule_] :=
  rawMatrix // ChangeDimension[#, 4] & // ExpandScalarProduct //
    ReplaceAll[#, rule] & // ReplaceAll[#, elimRule] & //
      Simplify;

(*---Process 4: \[Chi] \[Chi]bar->f fbar (General Charged Fermion)---*)
rule4 = GetMandelstamRules[s, t, u, mchi, mchi, mf, mf];
M4 = (gchi*gf)/denS[s]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]*
   SpinorUBar[p3, mf] . GA[\[Mu]] . SpinorV[p4, mf];
M4sqRaw =
```

```
  FermionSpinSum[M4*ComplexConjugate[M4]]/4 // Contract //
   DiracSimplify;
M4sq = ProcessCleanUp[M4sqRaw, rule4, {}];

(*---Process 5: \[Chi] \[Chi]bar->\[Nu] \[Nu]bar (General
   Neutrino)---*)
rule5 = GetMandelstamRules[s, t, u, mchi, mchi, 0, 0];
M5 = (gchi*gnu)/denS[s]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]*
   SpinorUBar[p3, 0] . GA[\[Mu]] . ((1 - GA[5])/2) . SpinorV[p4,
      0];

M5sqRaw =
  FermionSpinSum[M5*ComplexConjugate[M5]]/4 // Contract //
   DiracSimplify;
M5sq = ProcessCleanUp[M5sqRaw, rule5, {}];

Print["Process 4 (\[Chi] \[Chi]bar -> f fbar) : ", M4sq];
Print["Process 5 (\[Chi] \[Chi]bar -> \[Nu] \[Nu]bar) : ", M5sq];
```

1.2 Calculated Results

Process 4 ($\chi\bar{\chi} \to f\bar{f}$) :

$$\overline{|\mathcal{M}|^2} = \frac{2g_\chi^2 g_f^2\left[2m_\chi^4 + 2\left(2m_f^2 + s - t - u\right)m_\chi^2 + 2m_f^4 + t^2 + u^2 + 2m_f^2(s - t - u)\right]}{(M_A^2 - s)^2} \tag{1}$$

Process 5 ($\chi\bar{\chi} \to \nu\bar{\nu}$) :

$$\overline{|\mathcal{M}|^2} = \frac{g_\chi^2 g_\nu^2\left[2m_\chi^4 + 2(s - t - u)m_\chi^2 + t^2 + u^2\right]}{(M_A^2 - s)^2} \tag{2}$$

Take $\chi \to e, f \to \mu, A' \to \gamma$

$$\overline{|\mathcal{M}_4|^2} = \frac{2g_\chi^2 g_\nu^2\left[t^2+u^2\right]}{s^2}$$

$$\overline{|\mathcal{M}_5|^2} = \frac{g_\chi^2 g_\nu^2\left[t^2+u^2\right]}{s^2}$$

With a $\frac{1}{2}$difference.

This also correspond with FeynCalc website result for $e^+e^- \to \mu^+\mu^-$

1.3 Cross Section Calculation and Ratio Verification

Calculate cross section and take the paper 2603.03444's limit

```
pin = Sqrt[s/4 - mchi^2];
pf = Sqrt[s/4 - mf^2];
pnu = Sqrt[s/4];

(* expand s, t, u with scattering angle \theta or z *)
ruleKinF = {t -> mchi^2 + mf^2 - s/2 + 2*pin*pf*z,
   u -> mchi^2 + mf^2 - s/2 - 2*pin*pf*z};
ruleKinNu = {t -> mchi^2 - s/2 + 2*pin*pnu*z,
   u -> mchi^2 - s/2 - 2*pin*pnu*z};

(* conditions in arXiv: 2603.03444 *)
paperConditions = {mf -> 0, gf -> g, gnu -> g};

M4sqKin = M4sq /. ruleKinF /. paperConditions // Simplify;
M5sqKin = M5sq /. ruleKinNu /. paperConditions // Simplify;

(* 2->2 sigma = Integral[ |pf|/|pi|*|M|^2/(64pi^2*Ecm),{z,-1,1}]
   *)
(* under mf->0 condition, phase space factors p_out/p_in equal,
   so ratio of sigma = ratio of integral *)
(* initial states same --> pi=pi; final states same under the
   paper's mf, mnu=0 limit --> pf=pf *)

Sigma4Prop = Integrate[M4sqKin, {z, -1, 1}] // Simplify;
Sigma5Prop = Integrate[M5sqKin, {z, -1, 1}] // Simplify;

Print["--- Cross Section Proportions ---"];
Print["Integral of |M_{ff}|^2 : ", Sigma4Prop];
Print["Integral of |M_{\[Nu]\[Nu]}|^2 : ", Sigma5Prop];

ratio = Sigma5Prop/Sigma4Prop // Simplify;

Print["--- Verification of arXiv:2603.03444 Eq(16) ---"];
Print["Ratio \[Sigma](\[Chi] \[Chi]bar -> \[Nu] \[Nu]bar) / \[
   Sigma](\[Chi] \[Chi]bar -> f fbar) = ", ratio];
```

1.4 Calculated Results

Use $\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2|\bar{p}_f|}{64\pi^2 E_{CM}^2|\bar{p}_i|}$ Schwartz (5.32),

initial states same $\rightarrow p_i = p_i$;

final states same under the paper's $m_f, m_\nu = 0$ limit $\rightarrow p_f = p_f$

Integral for Charged Fermions ($\chi\bar{\chi} \rightarrow f\bar{f}$):

$$\int_{-1}^{1} |\mathcal{M}_{f\bar{f}}|^2 dz = \frac{8g^2 g_\chi^2 s\left(2m_\chi^2 + s\right)}{3\left(M_A^2 - s\right)^2} \tag{3}$$

Integral for Neutrinos ($\chi\bar{\chi} \to \nu\bar{\nu}$):

$$\int_{-1}^{1} |\mathcal{M}_{\nu\bar{\nu}}|^2 dz = \frac{4g^2 g_\chi^2 s\left(2m_\chi^2 + s\right)}{3\left(M_A^2 - s\right)^2} \tag{4}$$

Under $m_f \to 0$ and $m_\nu = 0$ limit $\frac{|p_f|}{|p_i|}$ are the sameσ ratio is the integral's ratio:

$$\frac{\sigma(\chi\bar{\chi} \to \nu\bar{\nu})}{\sigma(\chi\bar{\chi} \to f\bar{f})} = \frac{1}{2} \tag{5}$$

Which correspond to the equation (16) in paper 2603.03444

2 B3 B4 in 1709.07882

2.1 code

Listing 1: Mathematica code for SIDM constraints plot

```
ClearAll["Global`*"];

v = 10^-3;

limitB3[mx_, mA_] := 0.02*Sqrt[10^-3/mx]*(mA/1)^2;
limitB4[mx_] := 10^-10*(mx/1)^(3/2);

fixedMA = 10;
mchiR1 = fixedMA/v; (*line of R=1*)

plotMChi =
  LogLogPlot[{ConditionalExpression[limitB3[mx, fixedMA],
     mx <= mchiR1],
    ConditionalExpression[limitB3[mx, fixedMA], mx > mchiR1],
    ConditionalExpression[limitB4[mx], mx >= mchiR1],
    ConditionalExpression[limitB4[mx], mx < mchiR1]}, {mx, 10^-3,
    10^5},(*                    10^5          R=1          *)
     PlotRange -> {10^-16, 10^0},
   PlotStyle -> {Directive[Thick, Blue],
     Directive[Thick, Blue, Dashed, Opacity[0.6]],
     Directive[Thick, Red], Directive[Thick, Red, Dashed, Opacity
        [0.6]]},
    Filling -> {1 -> Top, 3 -> Top},
   FillingStyle -> Directive[Opacity[0.15], Black],
   GridLines -> {{mchiR1}, None},
   GridLinesStyle -> Directive[Thick, Gray, DotDashed],
   Epilog -> {Text[
      Style["R = 1", 14, Bold, Darker[Gray]], {mchiR1*0.7,
       10^-2}, {1,
      0}]}, Frame -> True,
   FrameLabel -> {"\!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\) [MeV]"
     ,
```

```
      "\!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)]\)"},
    PlotLabel ->
     StringForm[
      "Constraints vs \!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\) (
        Fixed \
\!\(\*SubscriptBox[\(M\), \(A\)]\) = `` MeV)", fixedMA],
    PlotLegends ->
     Placed[{LineLegend[{Directive[Thick, Blue],
         Directive[Thick, Red]}, {"Eq (B3): Heavy Mediator",
         "Eq (B4): Ultra-light"}],
       LineLegend[{Directive[Thick, Black],
         Directive[Thick, Black, Dashed]}, {"Asymptotic Regime",
         "Transition/Invalid Regime"}]}, Bottom],
    BaseStyle -> {FontSize -> 14, FontFamily -> "Helvetica"},
    ImageSize -> 450];


fixedMx = 10;
maR1 = fixedMx*v; (*R=1*)

plotMA =
  LogLogPlot[{ConditionalExpression[limitB3[fixedMx, mA], mA >=
     maR1],
     ConditionalExpression[limitB3[fixedMx, mA], mA < maR1],
    ConditionalExpression[limitB4[fixedMx], mA <= maR1],
    ConditionalExpression[limitB4[fixedMx], mA > maR1]}, {mA,
      10^-3,
    10^3}, PlotRange -> {10^-16, 10^0},
   PlotStyle -> {Directive[Thick, Blue],
     Directive[Thick, Blue, Dashed, Opacity[0.6]],
     Directive[Thick, Red], Directive[Thick, Red, Dashed, Opacity
        [0.6]]},
    Filling -> {1 -> Top, 3 -> Top},
   FillingStyle -> Directive[Opacity[0.15], Black],
   GridLines -> {{maR1}, None},
   GridLinesStyle -> Directive[Thick, Gray, DotDashed],
   Epilog -> {Text[
      Style["R = 1", 14, Bold, Darker[Gray]], {maR1*1.5, 10^-2},
        {-1,
       0}]}, Frame -> True,
   FrameLabel -> {"\!\(\*SubscriptBox[\(M\), \(A\)]\) [MeV]",
     "\!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)]\)"},
   PlotLabel ->
    StringForm[
     "Constraints vs \!\(\*SubscriptBox[\(M\), \(A\)]\) (Fixed \
\!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\) = `` MeV)", fixedMx],
   PlotLegends ->
    Placed[{LineLegend[{Directive[Thick, Blue],
        Directive[Thick, Red]}, {"Eq (B3): Heavy Mediator",
        "Eq (B4): Ultra-light"}],
      LineLegend[{Directive[Thick, Black],
```

```
        Directive[Thick, Black, Dashed]}, {"Asymptotic Regime",
        "Transition/Invalid Regime"}]}, Bottom],
   BaseStyle -> {FontSize -> 14, FontFamily -> "Helvetica"},
   ImageSize -> 450];

GraphicsRow[{plotMChi, plotMA}, ImageSize -> 950]
```

2.2 plot

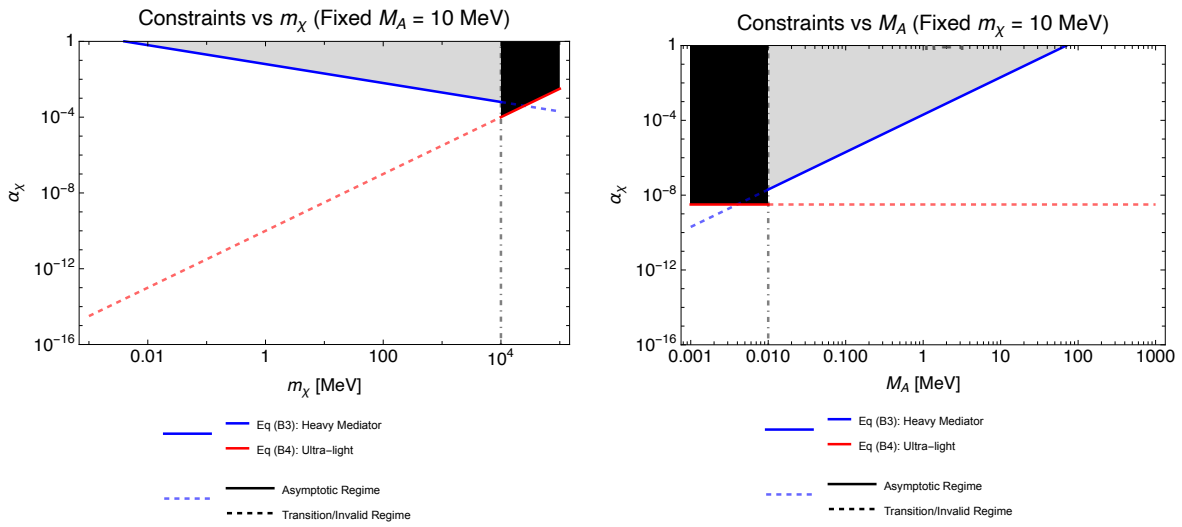


Figure 1: Self-interaction constraints on the coupling constant α_χ as a function of dark matter mass m_χ (left) and mediator mass M_A (right).

Effective area are far away from R=1 line.

3 B3, B4 & my result

3.1 code

Listing 2: Mathematica code for SIDM constraints plot

```
ClearAll["Global`*"]


limit = 4578.4;        (*GeV^-3*)
v = 10^-3;
mchi = 0.01;           (*10 MeV=0.01 GeV*)


alphaXDerived[R_] :=
  Sqrt[(limit*mchi^3*v^4)/(4*Pi*
      R^2*(R^2/(R^2 + 1) + Log[1/(R^2 + 1)]/(R^2 + 2)))];

alphaXB3[R_] := 0.02*Sqrt[10^-6/mchi]*((mchi*v)/(R*10^-3))^2;
```

```
alphaXB4[R_] := 10^-10*(mchi/10^-3)^(3/2);



(*Slope Verification)*)

(*slope=d(lnalpha)/d(lnR)=(1/alpha dalpha)/(1/R dR)=R/alpha*
   dalpha/dR*)

derivedSlope[R_] =
  Simplify[(R*D[alphaXDerived[R], R])/alphaXDerived[R]];
b3Slope[R_] = Simplify[(R*D[alphaXB3[R], R])/alphaXB3[R]];
b4Slope[R_] = Simplify[(R*D[alphaXB4[R], R])/alphaXB4[R]];


Print["R -> 0   my   result slpoe: ", Limit[derivedSlope[R], R ->
   0]];
Print["R -> 0 B3 slope: ", Limit[b3Slope[R], R -> 0]];

Print["\n R -> Infinity, my result slope: ",
  Limit[derivedSlope[R], R -> Infinity]];
Print["R -> Infinity, B4 slope: ", Limit[b4Slope[R], R ->
   Infinity]];







LogLogPlot[{alphaXDerived[R], alphaXB3[R], alphaXB4[R]}, {R,
   10^-3,
  10^3}, PlotStyle -> {Directive[Thick, Blue],
   Directive[Thick, Dashed, DarkGreen],
   Directive[Thick, DotDashed, Red]      }, Frame -> True,
 FrameLabel -> {Style[
    "Dimensionless Parameter R (v * \!\(\*SubscriptBox[\(m\), \
\(\[Chi]\)]\) / \!\(\*SubscriptBox[\(m\), \(\[Phi]\)]\))", 12,
   Bold],
   Style["Coupling Constant \!\(\*SubscriptBox[\(\[Alpha]\), \(\[
      Chi]\)]\)",
     12, Bold]}, GridLines -> Automatic,
 Filling -> {1 -> Top, 2 -> Top, 3 -> Top},
 FillingStyle -> Directive[Gray, Opacity[0.05]],
 PlotLegends ->
  Placed[LineLegend[{"My Derivation (\!\(\*SubscriptBox[\(\[Sigma
     ]\), \
\(T2\)]\))", "arXiv B3 Bound (R << 1)",
     "arXiv B4 Bound (R \[GrateSUpEqual] 1)"},
    LegendStyle -> Directive[FontSize -> 11]], {0.65, 0.75} ],
 PlotLabel ->
  Style["SIDM Bounds & Asymptotic Behavior Verification", 14,
```

```
      Bold],
58  ImageSize -> 550]
```

3.2 output

$R \to 0$, my result slope: -2.

$R \to 0$, B3 slope: -2.

$R \to \infty$, my result slope: -1.

$R \to \infty$, B4 slope: 0.

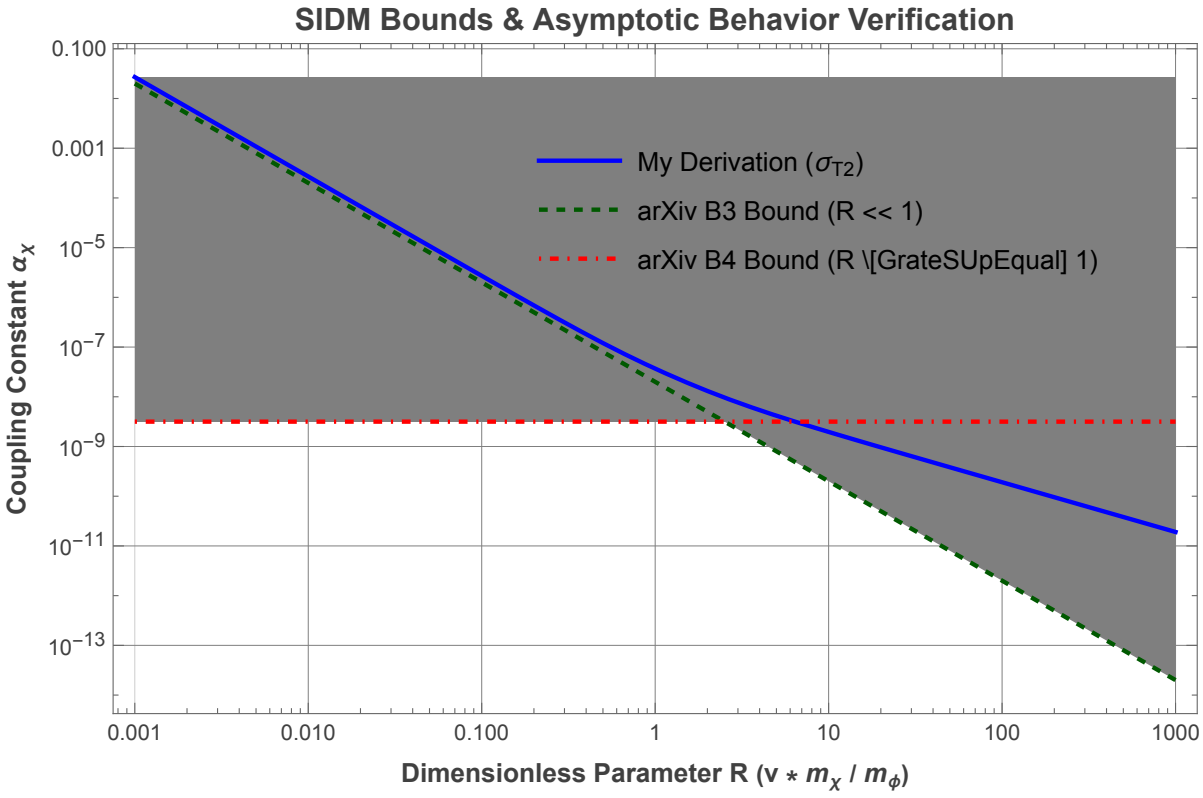


Figure 2

m_χ&v are constants, only M_A is variable, varying with R.

When R goes to ∞, my result will decrease by $\ln(\frac{1}{R})$ but B4 is not decreasing. B4 is not a function of M_A. It is not like what I got directly from classical calculation. There may be some other concerns when they got B4. I guess when $R \gg 1$, there might be some divergence that people have to drop, and cut down at some point. In our graph when R oes to ∞, the ratio will goes to ∞. Which means σ_{T2} diverges. The cut-off point might be the intersection of my result (Blue line) and B4 (Red dashed line) at $R \gg 1$. B4 makes sense when $R >\cong 1$, my result works only at $R \gg 1$ When $R \ll 1$, my result and B3 both works.

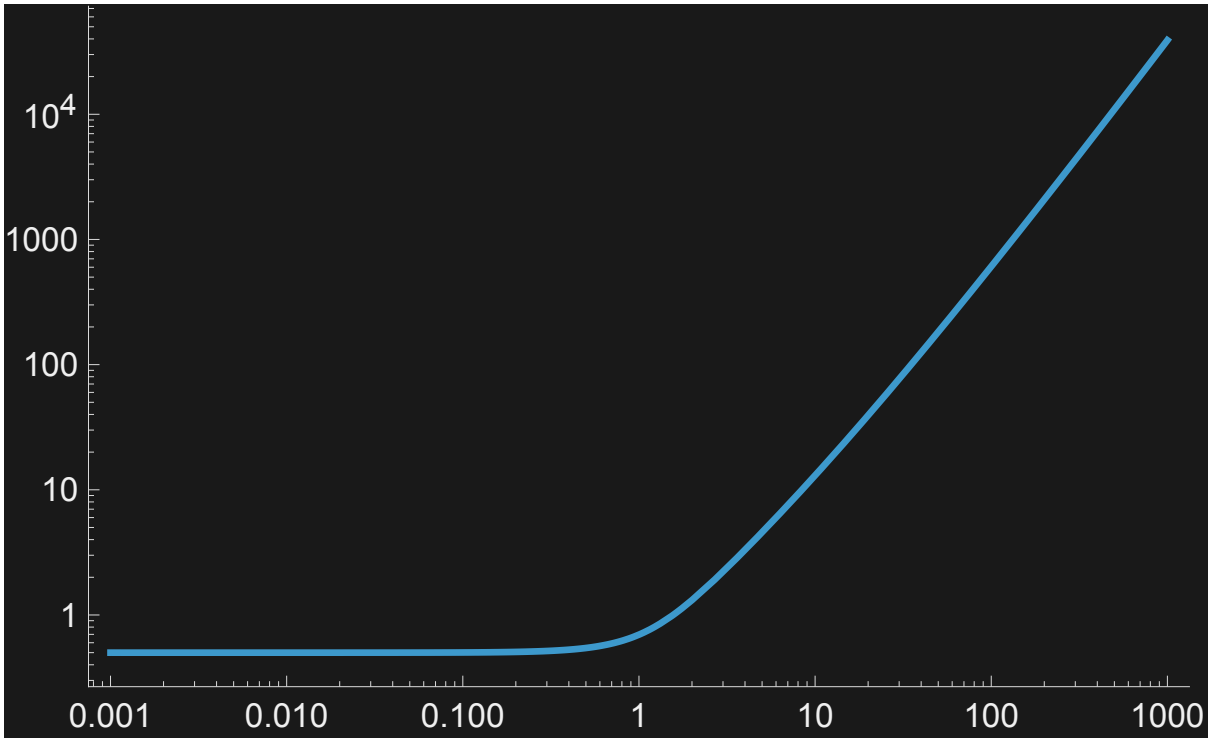


Figure 3